



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION JUNE 2024

Course : EC3283 (Cyber Security)
Lecturer : Muhammad Hamizan Johari

Time: 8.00 am – 11.00 am (3 hours)
Date : 30 September 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Briefly describe the following key security concepts.

i. Availability (2 MARKS)

ii. Confidentiality (2 MARKS)

b. Discuss the following computer security terminology.

i. Risk (3 MARKS)

ii. Attack (3 MARKS)

iii. Counter Measure (3 MARKS)

c. A bank transaction's information is an example that requires a strong encryption method for security. Implement and describe the steps required for creating and opening digital envelopes in the context of a bank transaction. (12 MARKS)

2. a. Briefly explain the following password cracking attacks.

i. Dictionary attack (3 MARKS)

ii. Rainbow table attack (3 MARKS)

b. Briefly discuss the access management of the ICAM approach. (4 MARKS)

c. Apply the following requirements for create **THREE (3)** different passwords. (6 MARKS)

i. Contains lowercase and uppercase letter.

ii. Contains at least **TWO (2)** numbers.

iii. Contains at least **ONE (1)** symbol character.

- d. A military intelligence unit, which is part of a larger defence organization, collects, analyses, and disseminates sensitive information about ongoing operations in a war zone. This unit handles a variety of sensitive data, including as satellite photography, intercepted conversations, and human intelligence reports. The intelligence gathered is critical for planning and carrying out military operations. (9 MARKS)

Given there are three high-ranking officers (a general, a captain and a sergeant) in the unit, assign the officers to the following files with the appropriate read, write and execute accesses.

- i. Confidential files.
- ii. Top secret files.
- iii. Secret files.

3. a. Describe ways on how to defend oneself against investment frauds. (6 MARKS)

- b. Briefly explain any **TWO (2)** components of a virus. (4 MARKS)

- c. Nowadays, social media plays a significant role in students' lives. Students rely on social media as a primary mode of communication with their lecturers and classmates. However, hackers also use social media as a means to generate income. By hacking into social media accounts, they can profit by blackmailing the account owners or by spreading false rumors about them online. This act is commonly referred to as identity theft. (6 MARKS)

As a frequent user of social media, plan how you can protect yourself from identity theft.

- d. Identify and describe the malware that could be attacking in each of the following scenario.

- i. Sam opens an unknown file attached to an email sent by his friend. Suddenly, his computer becomes infected, causing files to become corrupted. (3 MARKS)

- ii. Indica downloads an illegal game from an unknown website. Soon after, her computer files are secretly uploaded to a hacker's website without her knowledge. (3 MARKS)

- iii. A private company's database was suddenly filled to capacity during the night. No workers were present in the building at the time, and the database needed to be replaced with a new database. (3 MARKS)
4. a. Encrypt the plaintext, "MYNAMEISJEFF" using the given encryption techniques.
- i. Caesar Cypher (shift of 3) (3 MARKS)
- iii. Vignere Cypher (keyword: KEYKEYKEYKEY) (6 MARKS)
- b. A criminal has been caught, and the prosecutor requires evidence to be extracted from his mobile phone. Apply the mobile forensic techniques to obtain the evidence. Then, describe at least **ONE (1)** possible challenge an investigator could face during the extraction process. (8 MARKS)
- c. The TGV theatrical company wants to collect customer data through an online questionnaire on their website. However, the company wants to establish several terms and conditions before uploading the questionnaire online. (8 MARKS)
- Implement **FOUR (4)** key privacy aspects as requirements for the company's terms and conditions.

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